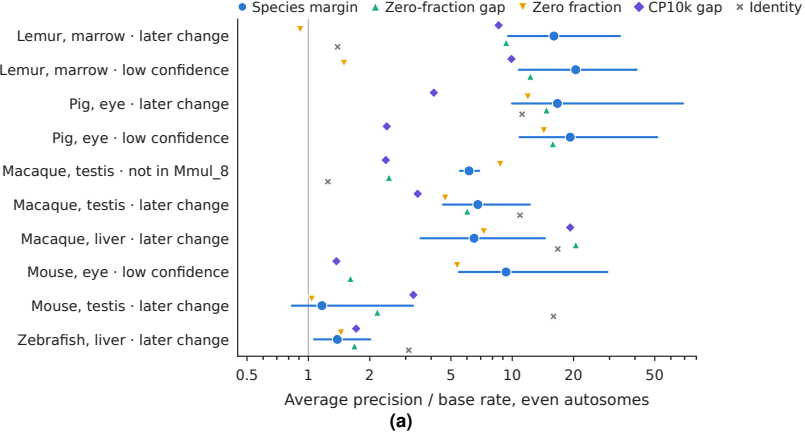
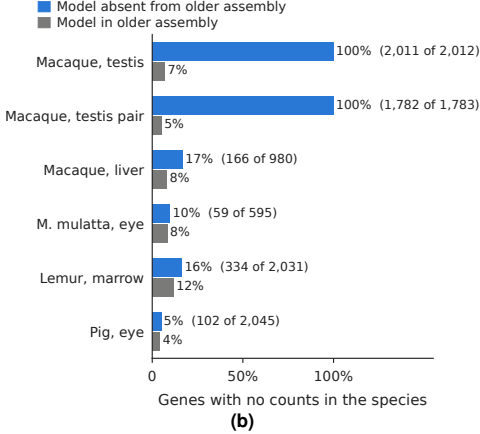
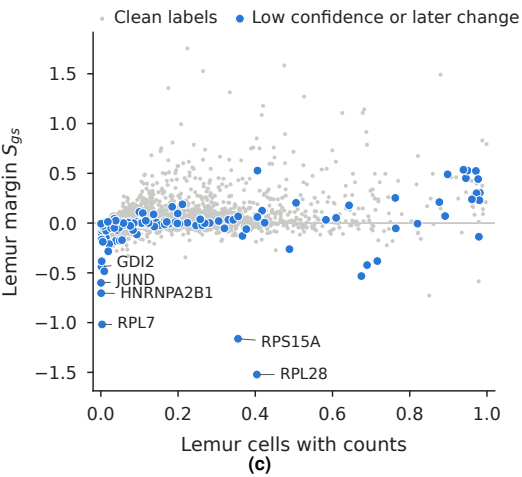

Species margin
Zero-fraction gap
Zero fraction
CP10k gap
Identity
Lemur, marrow · later change
Lemur, marrow · low confidence
Pig, eye · later change
Pig, eye · low confidence
Macaque, testis · not in Mmul_8
Macaque, testis · later change
Macaque, liver · later change
Mouse, eye · low confidence
Mouse, testis · later change
Zebrafish, liver · later change
0.5
1
2
5
10
20
50
Average precision / base rate, even autosomes

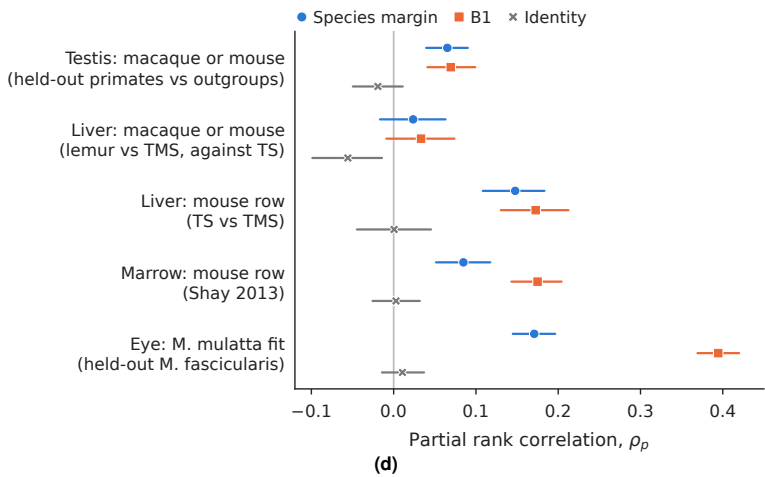
(a)


Model absent from older assembly
Model in older assembly
Macaque, testis
100% (2,011 of 2,012)
7%
Macaque, testis pair
100% (1,782 of 1,783)
5%
Macaque, liver
17% (166 of 980)
8%
M. mulatta, eye
10% (59 of 595)
8%
Lemur, marrow
16% (334 of 2,031)
12%
Pig, eye
5% (102 of 2,045)
4%
0
50%
100%
Genes with no counts in the species

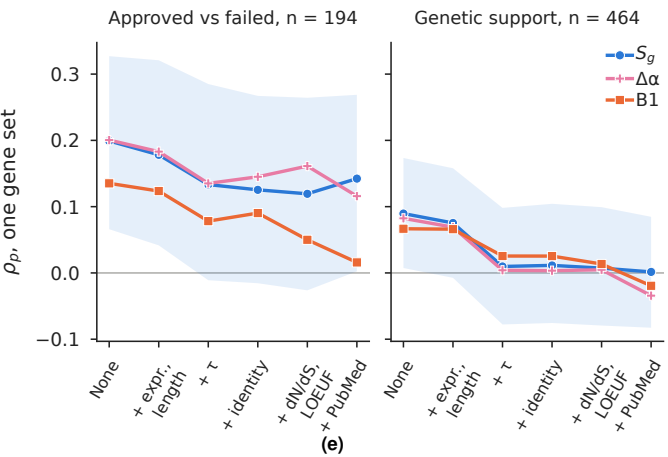
(b)


Clean labels
Low confidence or later change
Lemur margin S_{gs}
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
GDI2
JUND
HNRNPA2B1
RPL7
RPS15A
RPL28
0.0
0.2
0.4
0.6
0.8
1.0
Lemur cells with counts

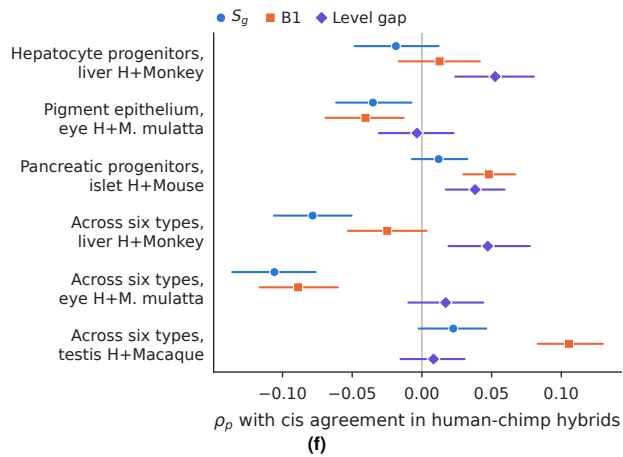
(c)


Species margin
B1
Identity
Testis: macaque or mouse
(held-out primates vs outgroups)
Liver: macaque or mouse
(lemur vs TMS, against TS)
Liver: mouse row
(TS vs TMS)
Marrow: mouse row
(Shay 2013)
Eye: M. mulatta fit
(held-out M. fascicularis)
−0.1
0.0
0.1
0.2
0.3
0.4
Partial rank correlation, ρ_p

(d)


Approved vs failed, n = 194
Genetic support, n = 464
S_g
Δα
B1
ρ_p, one gene set
0.3
0.2
0.1
0.0
−0.1
None
+ expr., length
+ τ
+ identity
+ dN/dS, LOEUF
+ PubMed

(e)


S_g
B1
Level gap
Hepatocyte progenitors,
liver H+Monkey
Pigment epithelium,
eye H+M. mulatta
Pancreatic progenitors,
islet H+Mouse
Across six types,
liver H+Monkey
Across six types,
eye H+M. mulatta
Across six types,
testis H+Macaque
−0.10
−0.05
0.00
0.05
0.10
ρ_p with cis agreement in human-chimp hybrids

(f)